

EEE3000X Report



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Henry Fagan

(a) Objective

I spent a total of 4 weeks interning at Henry Fagan. I worked in the electrical engineering department, assisting them with their consulting projects at the time. The main objective of this internship was to gain exposure to the software and site-verification processes used in professional electrical services design.

(b) Description of the equipment and principles involved

My work centred on two design software tools commonly used in the electrical consulting industry: AutoCAD, for creating and amending electrical layout drawings, and Relux, a lighting-simulation package used to verify that a proposed luminaire layout meets the illuminance (lux) level required for a given space. To do the drawings and lux simulations I had to verify that all proposed designs complied with SANS electrical standards (SANS 10142-1). This involved me reading the SANS book which outlined the required minimum lux levels for each type of room (eg bathroom, office, bedroom etc) and doing wiring size calculations as stated according to the regulations.

(c) Implementation

- Performed lux levels simulations using the Relux software. This involved determining the lux level required to meet the SANS standard for each individual room type. Then choosing the appropriate type of light fitting for each space and doing the simulation and adjusting the luminaire layouts as required.
- I performed calculations using the guidelines provided in SANS to determine the appropriate cable size for different purposes (eg wifi, alarm, lights etc). I also reviewed existing wiring layouts against the applicable SANS standards, checking for any deviations in aspects such as cable sizing and circuit protection.
- Updated ongoing electrical services projects in AutoCAD, amending layout and wiring drawings (such as cable routing, fixture layouts and alarm systems) to reflect design changes requested by the engineers on the team.
- Performed basic on-site checks. I visited project sites to confirm that installed electrical work matched the approved design drawings. Through these checks I got to see the whole project pipeline, and the management required from an engineer to ensure their design finishes to the standard as it was drawn.
- Edited Word and Excel documents in support of ongoing projects, including updating specifications, schedules, and project documentation. Made phone calls to light suppliers to inquire about pricing and product availability.

- Made a powerpoint presentation at the end of the internship detailing what I did and what I learnt during my time at Henry Fagan. I then presented it to the whole office at the end of my internship.

(d) Results obtained

By the end of my internship I was able to perform Relux simulations from start to finish and construct and edit AutoCAD drawings. Reviewing wiring layouts against SANS standards helped me identify aspects of designs that needed correction and my site checks also allowed me to see how to effectively check and ensure that the build is in line with the drawings.

(e) Conclusion

This internship gave me exposure to the consulting engineering profession. Apart from learning the technical side of the work, I was also able to join meetings with clients and regular staff meetings which allowed me to see how multi-disciplinary teams work together to deliver a complete project to a client. This opened my eyes to the professionalism and collaborative side of the job. Through this internship I also gained a deeper understanding of why SANS is required as I was able to place it in context and use it, as opposed to an arbitrary thing as it is mentioned in my coursework. The internship was rewarding as I got to see my lighting layouts actually being installed in a build which made me excited for the future once I can plan a complete project of my own.

CHPC Student Cluster Competition (National Round)

(a) Objective

I competed in this high-performance computing (HPC) competition in a team of four (Team EEElite). This competition consisted of designing a HPC cluster (made of nodes), and physically assembling, performing & optimising benchmarks on said cluster. The competition took place over 5 days. In this time the cluster needed to be assembled, software set-up, benchmarks run and optimised. My personal contribution to the team involved hardware design, cluster set-up, configuring remote access via ZeroTier, and running the MATLAB Monte Carlo Pi benchmark challenge.

(b) Description of the equipment and principles involved

A budget of R400 000 and a list of available nodes with varying specifications was provided. My knowledge of CPU, storage, memory, cabling and networking was needed to decide which nodes would be the best fit for the competition purposes. The operating system installed on the cluster was Rocky Linux 9.4, so my terminal command knowledge from the computer science courses was used to set up and operate the nodes.

The final cluster design consisted of:

- Head node: HPE ProLiant DL20 Gen11 (single socket), Intel Xeon E-2436 (6 cores / 12 threads, 2.9GHz), 16GB DDR5 RAM, 960GB SATA SSD.
- Compute nodes (x2): HPE ProLiant DL360 Gen11 (dual socket), 2 x Intel Xeon Silver 4510 (12 cores / 24 threads each, 2.4GHz), 48GB DDR5 RAM, 480GB SATA SSD.
- Networking: HPE Aruba CX 6100 12G switch with 10G SFP+ direct-attach copper cables linking all three nodes in a star topology.

(c) Implementation

The first step I did was to connect all the nodes together through the switch. A switch was chosen as it allowed for easier setup and connectivity between all the nodes. The OS was installed by plugging in a USB (which had Rocky Linux minimal ISO file on it) into each node and following the setup instructions. Once Rocky Linux was installed on all nodes the next step was configuring static networking on each node. This involved assigning the head and compute nodes fixed IP addresses on our cluster subnet with a shared gateway and DNS.

To set up the cluster the following low-level system configuration was done by me:

- Network File System (NFS) for a shared home directory. This involved exporting /home from the head node and mounting it on the compute nodes, so that user files and compiled binaries were available across the cluster.
- Configured ZeroTier to give our team secure remote access to the cluster. This allowed us to monitor and control the nodes from our own laptops during the competition. I set the head node up as the controlling point of our ZeroTier network: I created a private ZeroTier network, installed the ZeroTier client on the head node and on each team member's laptop, and joined each device to the network using its network ID. I then authorised each joined device from the ZeroTier network controller and assigned it an address on our private ZeroTier subnet. This gave us secure SSH access into the head node (and, from there, the compute nodes) over the cluster's private network from our own laptops, without needing to open ports directly on the head node's public-facing firewall.

The primary benchmark I was responsible for was the MATLAB Monte Carlo Pi challenge. This involved installing MATLAB across the cluster. I did this by downloading and installing MATLAB, along with the Parallel Computing Toolbox, onto each node across the cluster. I used our shared NFS home directory to make the installation and license files accessible from every node rather than repeating the install manually on each one. This saved a lot of time during the competition allowing me to spend more time optimizing the results. The main task for this benchmark was to optimize and parallelize MATLAB code to improve the execution speed of a Monte Carlo simulation

that is used to estimate the value of Pi. To do this I altered the provided MATLAB script (monteCarloPi_parallel.m) by tuning how the Parallel Computing Toolbox is used. This involved using a parfor loop (parallel for loop), and experimenting with different parpool configurations (Threads versus Processes versus local) to find the configuration that gave the best throughput on our node's core count. Having used Matlab in university it made me more comfortable to take on this challenge since I was working in a familiar environment.

(d) Results obtained

Our cluster ran the full suite of competition benchmarks (including HPCC, HPCG, MILC, the MATLAB Coding Challenge, LAMMPS, NWChem, and PennyLane) on the competition floor, with our team optimising the configuration in real time to ensure the best results. For the MATLAB Monte Carlo Pi challenge specifically, tuning the number of parallel workers and chunk size to match our node's core count measurably improved the sample throughput (millions of samples processed per second) compared to the default single-threaded configuration. Apart from the technical aspects, the competition was also an important lesson in the collaborative side of engineering. Being team leader, I learnt how to delegate and check in on the other members of my team to ensure all benchworks were progressing smoothly.

(e) Conclusion

The National Round gave me hands-on experience across the full stack of a real HPC cluster: contributing to hardware/software design under a fixed budget, low-level Linux system administration (networking, NFS, dependencies), secure remote access configuration with ZeroTier, and parallel-programming optimisation through the MATLAB Monte Carlo Pi challenge. This work drew directly on my computer engineering coursework in networking, computer architecture, and parallel/distributed computing. The competition gave me practical experience of the trade-offs involved in taking a design from budget-constrained specification through to a working, benchmarked cluster. The teamwork aspect also allowed me to take my knowledge learned from my professionalism courses and apply it. Providing an anchor for the principles learned in those courses, turning them into something I can relate to and actually visualise why they are so necessary for a project's success.

Overall Conclusion

Together, these two placements gave me more than six weeks of hands-on engineering experience, spanning electrical services design and site verification at Henry Fagan, and high-performance computing, networking, and parallel programming through the CHPC Student Cluster Competition. Both experiences reinforced concepts from my coursework and have strengthened my technical and professional skills ahead of completing my degree.